# Relative Entropy Does Not Supply an Omitted Finite-Central Calibration

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**Abstract**

Relative entropy and JLMS/OAQEC recovery comparisons are statements about the algebra and state family supplied to them. This note records the finite-center boundary. In a finite central decomposition with trace weights $\tau_a(c) = \sum_i a_i c_i$, the density of a central probability vector $p$ is $p_i/a_i$. Hence

$$S_{\tau_a}(p\|r) = \sum_i p_i \log \frac{p_i}{r_i},$$

so the common trace weights cancel from every pairwise relative entropy on a fixed retained face. The absolute entropy still contains the affine central term $\sum_i p_i \log a_i$. Therefore relative-entropy equality does not, by itself, determine an omitted finite central trace/area calibration or an absolute entropy-density representative. The positive exit is an affine or atomic calibration that fixes this central term on the state family, or a physical axiom excluding the hidden center.


**Status.** This is non-frozen external-review metadata for Team 1. It is intended as a small claim that a relative-entropy, OAQEC/JLMS, crossed-product, or black-hole-entropy reviewer can classify as known, false, missing-hypothesis, or useful. It does not modify any frozen Team 1 manuscript, Lean archive, public mirror manifest, or SHA-256 package.

# 1 Primary sources

The boundary below was checked against Araki's relative entropy for von Neumann algebras [1], the JLMS relative-entropy equality [2], entanglement-wedge reconstruction [3], Harlow's operator-algebra quantum error-correction formulation of the RT formula [4], and gravitational crossed-product entropy [5, 6]. These works are used as positive statements for supplied algebras, traces, and state families. No new relative-entropy theorem is claimed here.

## 2 Finite central face

Let

$$\mathcal{Z}_m = \ell^\infty(\{1, \ldots, m\}) \cong \mathbb{C}^m$$

and let $\tau_a(c) = \sum_i a_i c_i$, where $a_i > 0$. If $p$ is a central probability vector, its density with respect to $\tau_a$ is $p_i/a_i$. For two probability vectors $p, r$ with $\operatorname{supp} p \subseteq \operatorname{supp} r$,

$$S_{\tau_a}(p\|r) = \sum_{i:p_i>0} p_i \log \frac{(p_i/a_i)}{(r_i/a_i)} = \sum_{i:p_i>0} p_i \log \frac{p_i}{r_i}. \tag{1}$$

Thus the trace weights cancel from pairwise relative entropy on the retained central face.

The absolute entropy relative to the same trace is

$$H_{\tau_a}(p) = -\sum_i p_i \log \frac{p_i}{a_i} = H(p) + \sum_i p_i \log a_i. \tag{2}$$

The second term is an affine central calibration.

**Proposition 1** (relative entropy leaves a calibration fiber)**.** *Fix a retained central face $F \subseteq \{1, \ldots, m\}$ and a state family $P$ supported on $F$. If $a$ and $b$ are faithful trace calibrations on $F$, then every pairwise relative entropy among states in $P$ is the same for $\tau_a$ and $\tau_b$. Consequently pairwise relative entropy on $P$ does not determine the trace weights, the affine term $\sum_i p_i \log a_i$, or any absolute entropy-density coefficient depending on that term.*

*Proof.* Equation (1) contains only the probability ratios $p_i/r_i$. The weights $a_i$ cancel. Replacing $a$ by $b$ therefore leaves all pairwise relative entropies unchanged. Equation (2) shows that absolute entropy still depends on the affine functional $p \mapsto \sum_i p_i \log a_i$, which has not been fixed by those relative entropies. □

**Proposition 2** (two-atom witness)**.** *On the two-atom center, let $a = (1, 3)$, $b = (2, 2)$, and $p = (1/2, 1/2)$. The pairwise relative entropies on the two-atom probability simplex are unchanged when $\tau_a$ is replaced by $\tau_b$, but*

$$H_{\tau_a}(p) - H_{\tau_b}(p) = \frac{1}{2} \log \frac{3}{4} \neq 0.$$

*Proof.* The first statement is Proposition 1. For the second,

$$H_{\tau_a}(p) - H_{\tau_b}(p) = \frac{1}{2}(\log 1 + \log 3 - \log 2 - \log 2) = \frac{1}{2} \log \frac{3}{4}.$$

□

**Theorem 1** (affine central calibration criterion)**.** *Let $P$ be the retained central state family and write*

$$L_a(p) = \sum_i p_i \log a_i.$$

*An absolute central entropy or area calibration determines the finite-center entropy term on $P$ exactly when it determines the affine functional $L_a$ on the affine hull of $P$. In particular, vertex calibrations on every central atom recover all $\log a_i$ up to the common trace-normalization convention; calibrations on a proper face leave the omitted vertices unconstrained.*

*Proof.* By (2), all dependence on trace calibration is through the affine functional $L_a$. Knowing absolute entropy or area values on an affinely spanning subset of $P$ fixes $L_a$ on the affine hull of $P$, and no smaller calibration can fix it outside its affine span. Vertex tests read off the values $L_a(e_i) = \log a_i$. If a vertex is omitted, changing only that coordinate gives another faithful calibration with the same supplied tests and different absolute entropy for states using the omitted vertex. A common rescaling $a \mapsto \lambda a$ adds the constant $\log \lambda$ to all $L_a(p)$, which is invisible only when the intended convention is projective. □

# 3 JLMS/OAQEC boundary

JLMS and operator-algebra quantum error correction give powerful positive statements for the declared algebra and state family. They become a positive exit from this Team 1 obstruction exactly when the comparison includes the central face and its absolute affine calibration.

If the comparison factors through a quotient, selected central character, declared code algebra, or state family that omits some central vertices, then relative entropy remains algebra-relative. It does not certify maximality of the finite center or determine the omitted trace/area calibration.

# 4 Referee question

> In the target black-hole/AQFT reconstruction setting, does the invoked JLMS/OAQEC/crossed-product result supply full central-face affine calibration, or only relative entropy for the algebra and state family already specified?

If the answer is the second, Team 1 has no new relative-entropy theorem. The surviving claim is only the claim-boundary warning: relative entropy and recovery data should not be overread as finite-center maximality or as an absolute finite-center calibration principle.

# References


[1] H. Araki, *Relative entropy of states of von Neumann algebras*, Publ. RIMS 11 (1976), 809–833, EMS PDF.

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[3] X. Dong, D. Harlow, and A. C. Wall, *Reconstruction of Bulk Operators within the Entanglement Wedge in Gauge-Gravity Duality*, Phys. Rev. Lett. 117 (2016), 021601, arXiv:1601.05416.

[4] D. Harlow, *The Ryu-Takayanagi Formula from Quantum Error Correction*, Commun. Math. Phys. 354 (2017), 865–912, arXiv:1607.03901.

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